Eco Essential

Eco Essentials Data Warehousing

Enterprise Data Warehouse Design

1. Draft a bus matrix outlining key business processes.

Business Process	Dimensions	Fact Table
Sales Process	Date, Customer, Product, Promotional Campaign, Order	Fact_Sales
Marketing Email Conversion Process	Date, Customer, Email Campaign, Event Type	Fact_Email_Marketing_Events

2. Star Schema Creation
   A. Select the business process

      Sales Process: Tracking online product purchases and performance trends.

      Marketing Email Conversion Process: Tracking marketing emails sent to customers and their engagement or conversion outcomes.

   B. Declare the Grain

      Sales Process Grain: Each record in the sales fact table represents a single line item within an order.

      Marketing Email Conversion Grain: Each record in the marketing fact table represents a single email event sent to or interacted with by a customer.

C. Identify Dimensions

Sales Process Dimensions:

dim_customer: Contains customer information such as customer_ID, name, DOB, phone, email, address, city, state, zip.

dim_product: Includes product details such as product ID, name, type, supplier and price.

dim_promotional_campaign: Stores campaign ID, name, and discount amount.

dim_date: Captures timestamps for orders, allowing time-based analysis (day, month, year, quarter).

dim_order: Contains order-level attributes such as order ID and timestamp and payment type

Marketing Email Conversion Dimensions:

dim_customer: Shared across both processes, representing the same individuals.

dim_email_campaign: Contains email campaign attributes such as campaign ID, campaign name, and email ID.

dim_event_type: Defines the type of event, such as "Email Sent," and "Email Opened" .

dim_date: Represents both the send date and event timestamp.

D. Identify Facts

Fact Table for Sales Process: fact_sales
Foreign Keys: customer_id, product_id, campaign_id, date_id, order_id
Measures: quantity, total_sales

Fact Table for Marketing Email Conversion: fact_email_marketing_events

Foreign Keys: customer_id, campaign_id, event_type_id, date_id
Measures: email_sent_count, email_open_count, conversion_flag

3. Now combine the two separate star schemas into one enterprise data warehouse.

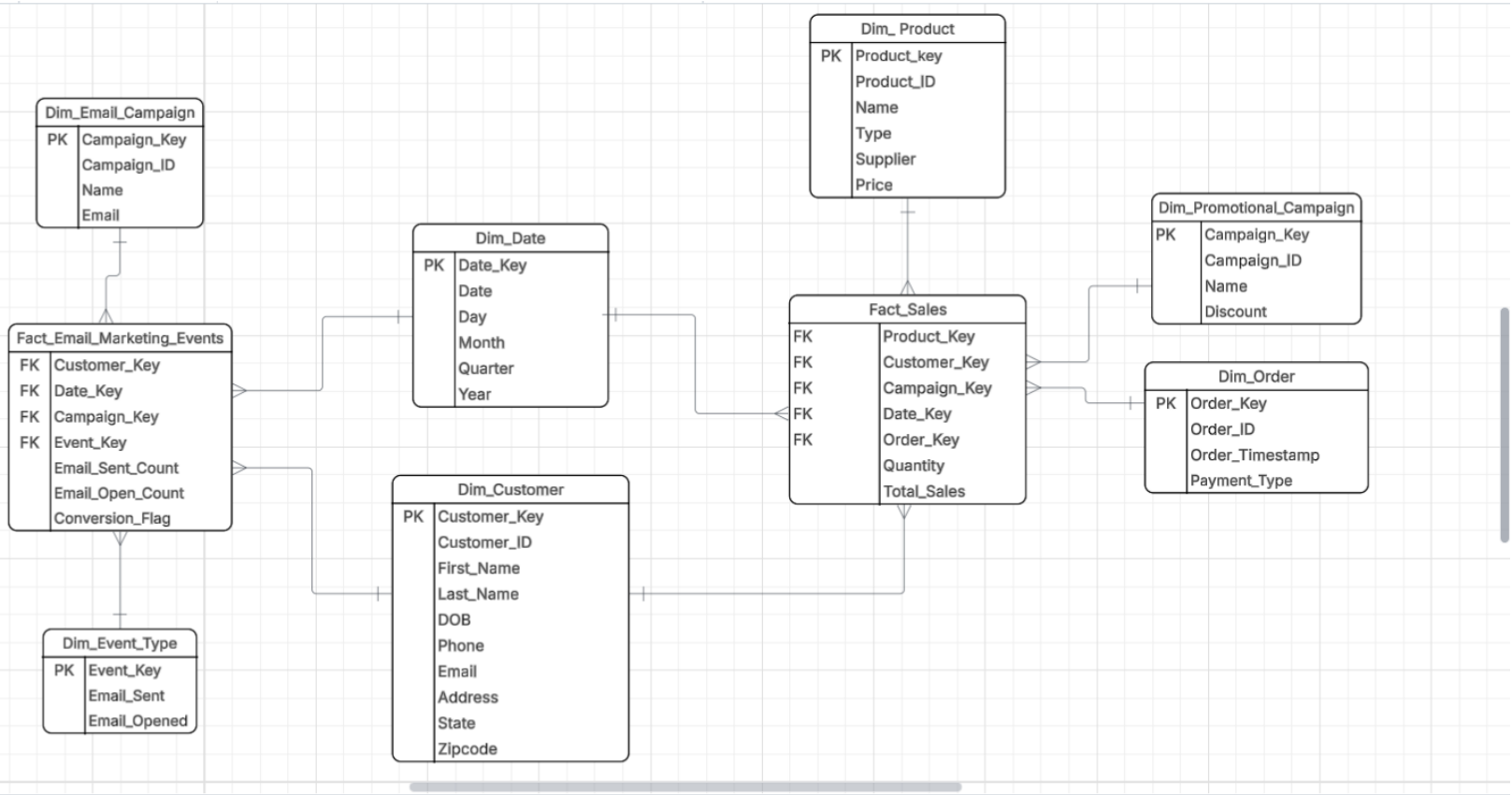
The Sales and Marketing data models wil be connected through the dim_customer and dim_date tables. This way analysts can correlate marketing activity (such as email opens or clicks) with subsequent purchases in the sales system.

4. Identify conformed dimensions to ensure consistency across schemas. You may want to highlight the conformed dimensions.

dim_customer.

dim_date

5. Submit (a legible) screenshot of your dimensional model in LucidChart. (Please label your objects with either dim_ or fact_)



53%

Search

12:42 PM
10/19/2025

Extract, Load, and Transform

While reflecting back on the first deliverable I decided to drop email campaign and event type tables. After loading the data, I changed the table columns according to the data I received.

The first thing I did next was create my src.yml.

_src_ecoessentials.yml

Code:

```
version: 2


sources:
  - name: ecoessentials_transactional
    database: CHRISTIANKERNS
    schema: ECOESSENTIALS_TRANSACTIONAL_DB
    tables:
      - name: customer
      - name: orders
        identifier: "\"ORDER\""
      - name: order_line
      - name: product
      - name: promotional_campaign


  - name: ecoessentials_marketing
    database: CHRISTIANKERNS
    schema: ECOESSENTIALS_MARKETING_EMAILS
    tables:
```

- name: marketing_emails

(This was a challenge in fixing later on, due to the word ORDER causing errors while creating dim_order. This was the solution I figured out.)

Then started to create my dim tables first with **Dim_Customer**

Code:

```
{{ config(materialized = 'table', schema = 'ecoessentials_dw') }}

SELECT
    CUSTOMER_ID AS customer_key,
    CUSTOMER_ID,
    CUSTOMER_FIRST_NAME AS first_name,
    CUSTOMER_LAST_NAME AS last_name,
    CUSTOMER_ADDRESS AS address,
    CUSTOMER_CITY AS city,
    CUSTOMER_STATE AS state,
    CUSTOMER_ZIP AS zip,
    CUSTOMER_COUNTRY AS country,
    CUSTOMER_PHONE AS phone
FROM {{ source('ecoessentials_transactional', 'customer') }}
```

Dim_Product

```
{{ config(materialized = 'table', schema = 'ecoessentials_dw') }}
```

```
SELECT
    PRODUCT_ID AS product_key,
    PRODUCT_ID,
    PRODUCT_NAME,
    PRODUCT_TYPE,
    PRICE
FROM {{ source('ecoessentials_transactional', 'product') }}
```

Dim_Order

```
{{ config(
    materialized = 'table',
    schema = 'ecoessentials_dw'
) }}

SELECT
    ORDER_ID AS order_key,
    ORDER_ID,
    CUSTOMER_ID,
    ORDER_TIMESTAMP
FROM {{ source('ecoessentials_transactional', 'orders') }}
```

Dim_Campaign

```
{{ config(materialized = 'table', schema = 'ecoessentials_dw') }}

SELECT
```

```
    CAMPAIGN_ID AS campaign_key,
    CAMPAIGN_ID,
    CAMPAIGN_NAME,
    CAMPAIGN_DISCOUNT
FROM {{ source('ecoessentials_transactional', 'promotional_campaign') }}
```

Dim_Date

```
{{ config(materialized = 'table', schema = 'ecoessentials_dw') }}

WITH cte_date AS (
    {{ dbt_date.get_date_dimension("1990-01-01", "2050-12-31") }}
)
SELECT
    date_day AS date_key,
    date_day,
    day_of_week,
    month_of_year,
    month_name,
    quarter_of_year,
    year_number
FROM cte_date
```

Then I started my facts

Fact_Sales

```
{{ config(materialized='table', schema='ecoessentials_dw') }}
```

```
SELECT
    p.product_key,
    c.customer_key,
    cm.campaign_key,
    d.date_key,
    o.order_key,
    ol.quantity,
    (ol.quantity * ol.price_after_discount) AS total_sales
FROM {{ source('ecoessentials_transactional', 'order_line') }} ol
INNER JOIN {{ ref('dim_order') }} o
    ON ol.ORDER_ID = o.ORDER_ID
INNER JOIN {{ ref('dim_product') }} p
    ON ol.PRODUCT_ID = p.PRODUCT_ID
INNER JOIN {{ ref('dim_customer') }} c
    ON o.CUSTOMER_ID = c.CUSTOMER_ID
LEFT JOIN {{ ref('dim_campaign') }} cm
    ON ol.CAMPAIGN_ID = cm.CAMPAIGN_ID
INNER JOIN {{ ref('dim_date') }} d
    ON d.date_key = CAST(o.ORDER_TIMESTAMP AS DATE)
```

Fact_Marketing

```
{{ config(materialized = 'table', schema = 'ecoessentials_dw') }}

WITH marketing_base AS (
```

```
    SELECT
        e.emaileventid AS email_event_key,
        e.emailid,
        e.campaignid,
        e.subscriberid AS customer_id,
        e.eventtype,
        e.eventtimestamp,
        CASE
            WHEN LOWER(e.eventtype) LIKE '%open%' THEN 1 ELSE 0
        END AS email_open_flag,
        CASE
            WHEN LOWER(e.eventtype) LIKE '%sent%' THEN 1 ELSE 0
        END AS email_sent_flag
    FROM {{ source('ecoessentials_marketing', 'marketing_emails') }} e
),

customer_conversion AS (
    SELECT
        c.customer_key,
        CASE
            WHEN COUNT(o.order_id) > 0 THEN 1
            ELSE 0
        END AS conversion_flag
    FROM {{ ref('dim_customer') }} c
    LEFT JOIN {{ source('ecoessentials_transactional', 'orders') }} o
        ON o.customer_id = c.customer_key
    GROUP BY c.customer_key
```

```
)

SELECT
    f.email_event_key,
    f.emailid,
    f.campaignid,
    f.customer_id,
    d.date_key,
    d.date_day,
    d.month_of_year,
    d.year_number,
    f.email_open_flag,
    f.email_sent_flag,
    cc.conversion_flag
FROM marketing_base f
LEFT JOIN {{ ref('dim_customer') }} c
    ON f.customer_id = c.customer_key
LEFT JOIN {{ ref('dim_date') }} d
    ON DATE(f.eventtimestamp) = d.date_day
LEFT JOIN customer_conversion cc
    ON f.customer_id = cc.customer_key
```

Data warehouse populated in Snowflake

app.snowflake.com/sfedu02/gpb20388/w3YdTCyusCYj#query

2025-10-12 12:41pm

Databases Worksheets

TRAINING_ROLE • ANIMAL_TASK_WH (X-Small) Share

Search objects

ECOESSENTIALS_DW
- Tables
  - DIM_CAMPAIGN
  - DIM_CUSTOMER
  - DIM_DATE
  - DIM_ORDER
  - DIM_PRODUCT
  - FACT_MARKETING
  - FACT_SALES
- ECOESSENTIALS_MARKETING_E...

FACT_MARKETING 73 Rows

EMAIL_EVENT_KEY	NUMBER(38,0)
EMAILID	NUMBER(38,0)
CAMPAIGNID	NUMBER(38,0)
CUSTOMER_ID	NUMBER(38,0)
DATE_KEY	DATE
DATE_DAY	DATE
MONTH_OF_YEAR	NUMBER(38,0)

No Database selected Settings Open in Workspaces Code Versions

```
1  select :datebucket(created), count(1) from table group by 1
```

Results Chart

Re-run your query. Query results are cached for 24 hours.

Query Details
Query duration 753ms
Rows 10
Query ID 01bff957-0106-c27a-0...

Ask Copilot

60°F Clear Search 5:51 PM 11/2/2025

Business Question 1

Which promotional campaign generated the most total sales?

app.snowflake.com/sfedu02/gpb20388/w3YdTCyusCYj#query

2025-10-12 12:41pm

Databases Worksheets

TRAINING_ROLE • ANIMAL_TASK_WH (X-Small) Share

Search objects

ECOESSENTIALS_DW
- Tables
  - DIM_CAMPAIGN
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  - DIM_DATE
  - DIM_ORDER
  - DIM_PRODUCT
  - FACT_MARKETING
  - FACT_SALES
- ECOESSENTIALS_MARKETING_E...

FACT_MARKETING 73 Rows

EMAIL_EVENT_KEY	NUMBER(38,0)
EMAILID	NUMBER(38,0)
CAMPAIGNID	NUMBER(38,0)
CUSTOMER_ID	NUMBER(38,0)
DATE_KEY	DATE
DATE_DAY	DATE
MONTH_OF_YEAR	NUMBER(38,0)

CHRISTIANKERNS.ECOESSENTIALS_DW Settings Open in Workspaces Code Versions

```
1  USE DATABASE CHRISTIANKERNS;
2  USE SCHEMA ECOESSENTIALS_DW;
3
4
5  SELECT c.campaign_name, SUM(s.total_sales) AS total_sales
6  FROM fact_sales s
7  JOIN dim_campaign c
8      ON s.campaign_key = c.campaign_key
9  GROUP BY c.campaign_name
10 ORDER BY total_sales DESC
11 LIMIT 5;
```

Results Chart

	CAMPAIGN_NAME	TOTAL_SALES
1	New Customer Welcome	1564.74
2	Newsletter Subscriber Special	1313.75
3	Kitchen Essentials Bundle	862.56
4	Holiday Gift Collection	774.56
5	Seasonal Clearance	596.98

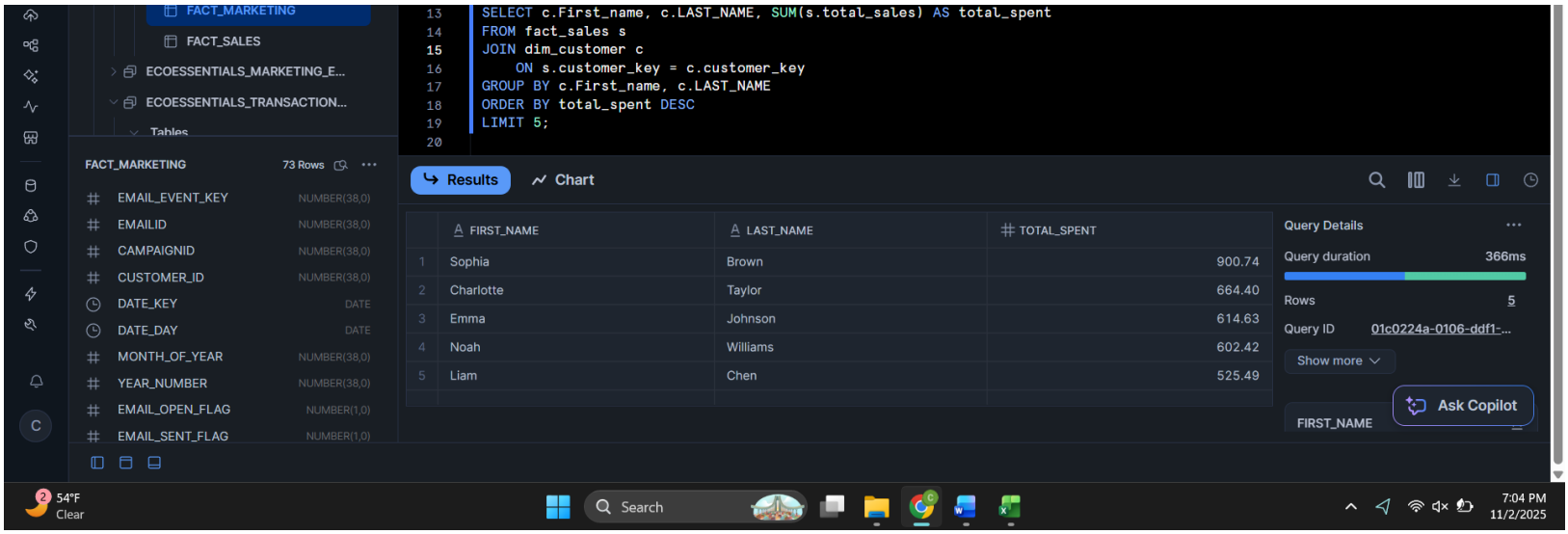
Query Details
Query duration 626ms
Rows 5
Query ID 01c02235-0106-ddf1-...
Show more

Ask Copilot

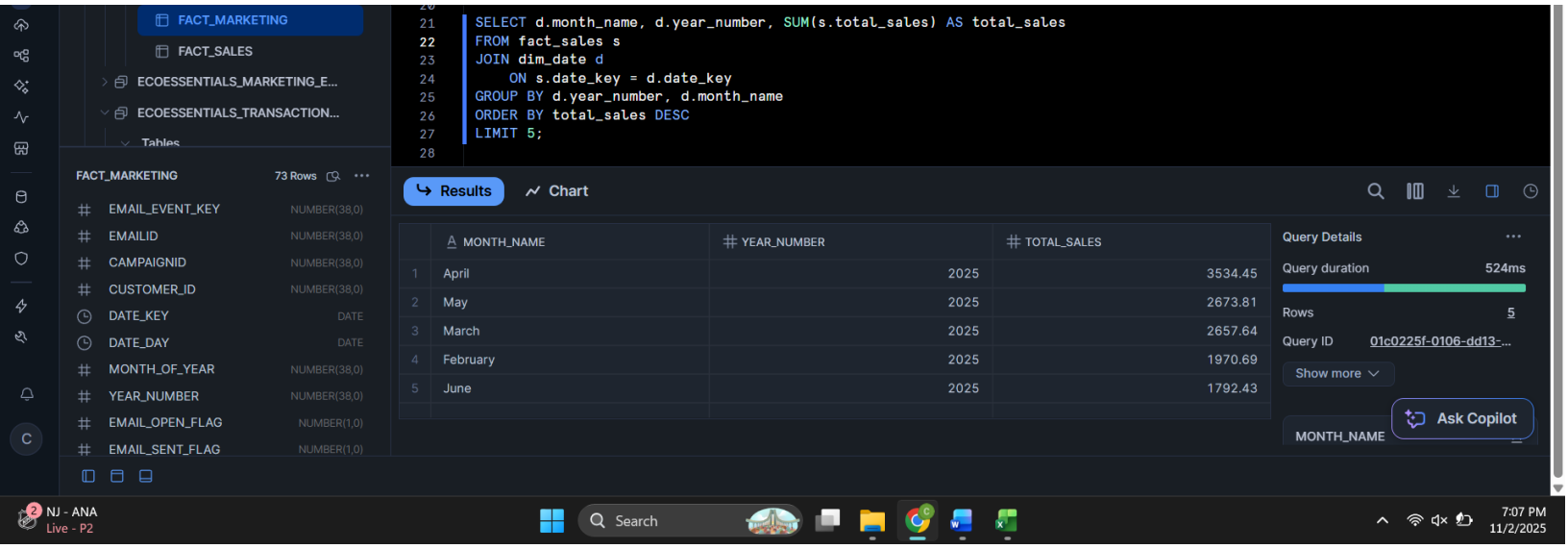
CAMPAIGN_NAME

54°F Clear Search 6:28 PM 11/2/2025

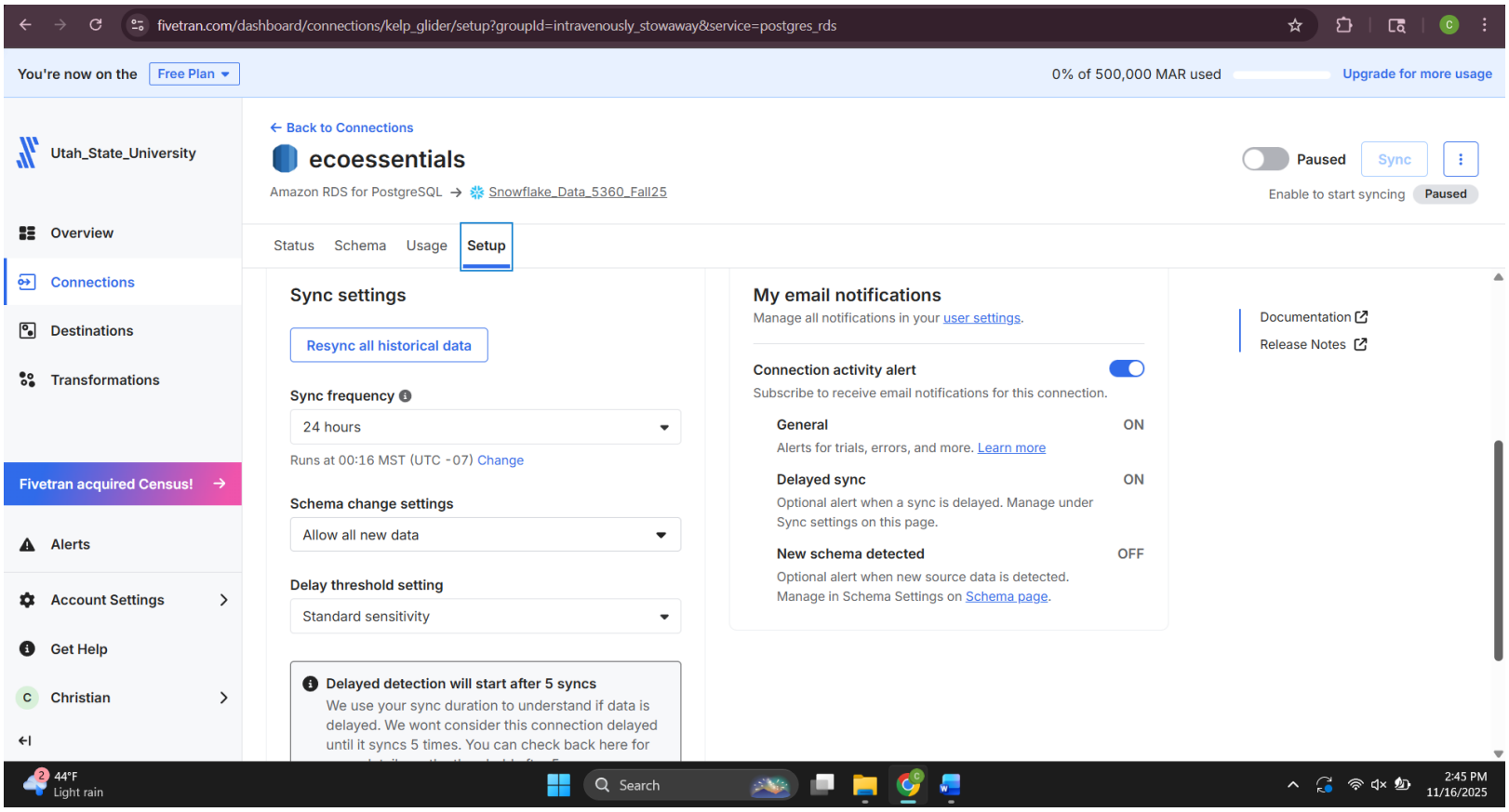
Which customers have spent the most money overall?



Which month had the highest total sales?



Testing & Scheduling



fivetran.com/dashboard/connections/fibre_throwback/setup?groupId=intravenously_stowaway&service=s3

You're now on the Free Plan

0% of 500,000 MAR used Upgrade for more usage

Utah_State_University

- Overview
- Connections
- Destinations
- Transformations

Fivetran acquired Census!

- Alerts
- Account Settings
- Get Help
- Christian

← Back to Connections

ecoessentials_marketing_emails.salesforce_marketing_emails

Amazon S3 → Snowflake_Data_5360_Fall25

Paused Sync

Enable to start syncing Paused

Status Schema Usage Setup

Sync settings

Resync all historical data

Sync frequency

24 hours

Runs at 00:16 MST (UTC -07) Change

Schema change settings

Allow all new data

Delay threshold setting

Standard sensitivity

Delayed detection will start after 5 syncs

We use your sync duration to understand if data is delayed. We wont consider this connection delayed until it syncs 5 times. You can check back here for

My email notifications

Manage all notifications in your user settings.

Connection activity alert

Subscribe to receive email notifications for this connection.

General ON

Alerts for trials, errors, and more. Learn more

Delayed sync ON

Optional alert when a sync is delayed. Manage under Sync settings on this page.

New schema detected OFF

Optional alert when new source data is detected. Manage in Schema Settings on Schema page.

Documentation

Release Notes

44°F Light rain

Search

2:46 PM 11/16/2025

hr575.us1.dbt.com/deploy/70471823500518/projects/70471823514772/jobs/70471823529999/settings

dbt

- Dashboard
- Studio
- Canvas
- Orchestration
- Documentation

Orchestration / Environments / Production PROD / Ecoessentials DW Refresh

Ecoessentials DW Refresh

Edit

Job settings

Job name

Ecoessentials DW Refresh

Consider choosing a name that's easily understood by your teammates.

+ Add description

Environment

Production

Execution settings

Commands

Run source freshness

Enables `dbt source freshness` as the first step of this job, without breaking subsequent steps. See the docs for more info.

```
1 dbt build -s ecoessentials.*
```

Generate docs on run

Automatically update dbt Catalog documentation each time this job runs

Schedule

Configure when this job should run.

Run on schedule

Triggers a run when the schedule is active.

Timing

Specific hours

Run at (UTC)

23

- Set up CLI
- Leave feedback
- Ask support assistant
- Get resources
- Utah State University
- Christian Kerns

45°F Rain

Search

2:56 PM 11/16/2025

hr575.us1.dbt.com/deploy/70471823500518/projects/70471823514772/runs/70471853692558

dbt

- Dashboard
- Studio
- Canvas
- Orchestration
- Documentation
- Set up CLI
- Leave feedback
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- Utah State University
- Christian Kerns

Prev Next

Rerun now

Invoke dbt with `dbt build -s ecoessentials.*`

Warning 15 seconds

System logs

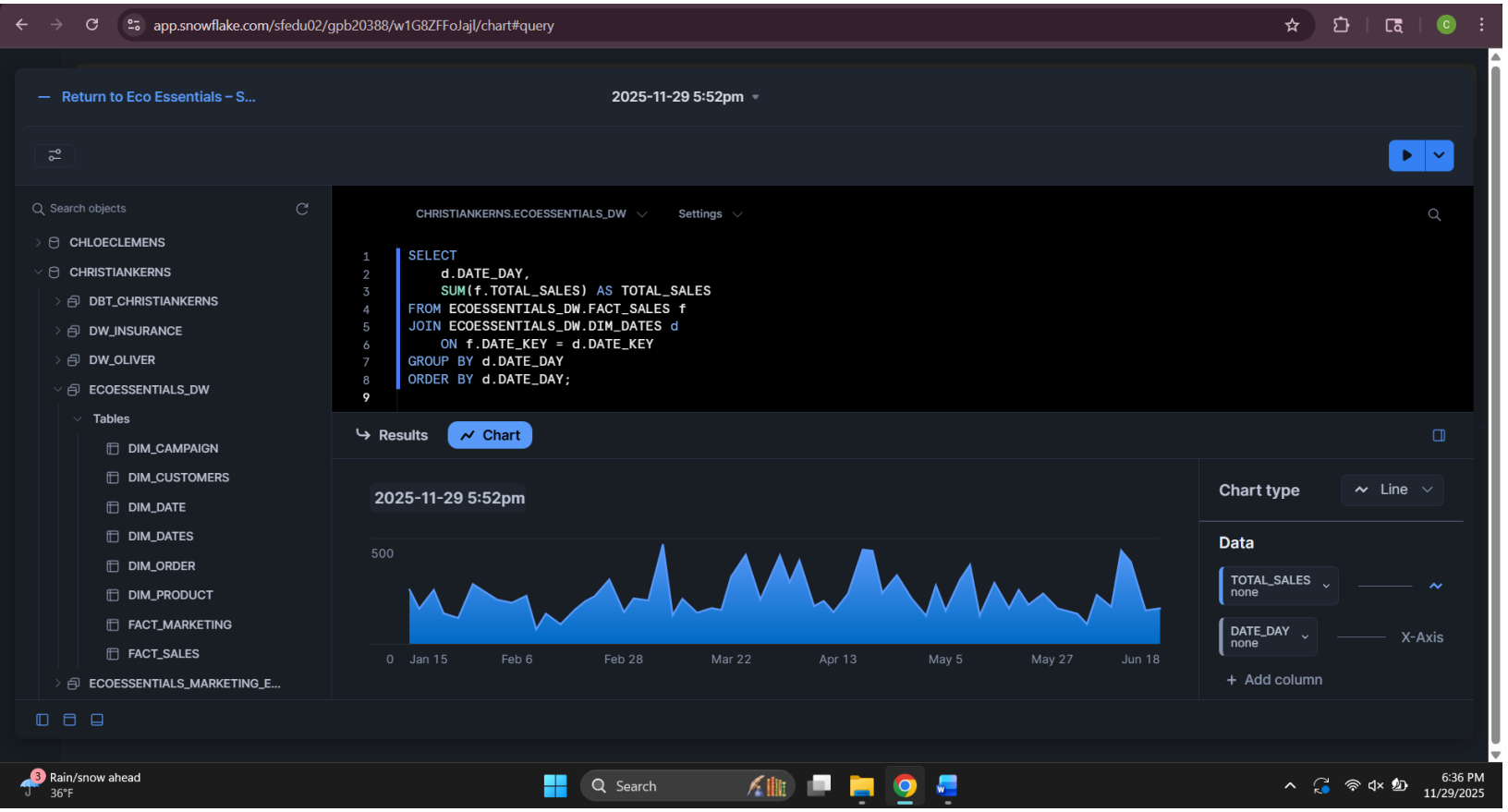
Search logs... 2 Warnings 1 Deprecation 14 Successes Debug logs Download

```
21:35:57 1 of 14 START sql table model ecoessentials_dw.dim_campaign ..................... [RUN]
21:35:57 2 of 14 START sql table model ecoessentials_dw.dim_customers ................... [RUN]
21:35:57 3 of 14 START sql table model ecoessentials_dw.dim_dates ....................... [RUN]
21:35:57 4 of 14 START sql table model ecoessentials_dw.dim_order ....................... [RUN]
21:35:58 4 of 14 OK created sql table model ecoessentials_dw.dim_order .................. [SUCCESS 1 in 1.12s]
21:35:58 5 of 14 START sql table model ecoessentials_dw.dim_product ..................... [RUN]
21:35:58 2 of 14 OK created sql table model ecoessentials_dw.dim_customers ............... [SUCCESS 1 in 1.16s]
21:35:58 6 of 14 START test unique_dim_order_order_key ................................ [RUN]
21:35:59 1 of 14 OK created sql table model ecoessentials_dw.dim_campaign ............... [SUCCESS 1 in 1.29s]
21:35:59 7 of 14 START test unique_dim_customers_customer_key ........................... [RUN]
21:35:59 6 of 14 PASS unique_dim_order_order_key ....................................... [PASS in 0.30s]
21:35:59 8 of 14 START test accepted_values_dim_campaign_campaign_discount__0_10__0_15__0_20__0_25__0_30__0_35  [RUN]
21:35:59 3 of 14 OK created sql table model ecoessentials_dw.dim_dates .................. [SUCCESS 1 in 1.69s]
21:35:59 9 of 14 START test not_null_dim_dates_date_key ............................... [RUN]
21:35:59 8 of 14 PASS accepted_values_dim_campaign_campaign_discount__0_10__0_15__0_20__0_25__0_30__0_35  [PASS in 0.33s]
21:35:59 7 of 14 PASS unique_dim_customers_customer_key ............................... [PASS in 0.59s]
21:35:59 9 of 14 PASS not_null_dim_dates_date_key ...................................... [PASS in 0.38s]
21:35:59 10 of 14 START sql table model ecoessentials_dw.fact_marketing ................ [RUN]
21:36:00 5 of 14 OK created sql table model ecoessentials_dw.dim_product ............... [SUCCESS 1 in 1.54s]
```

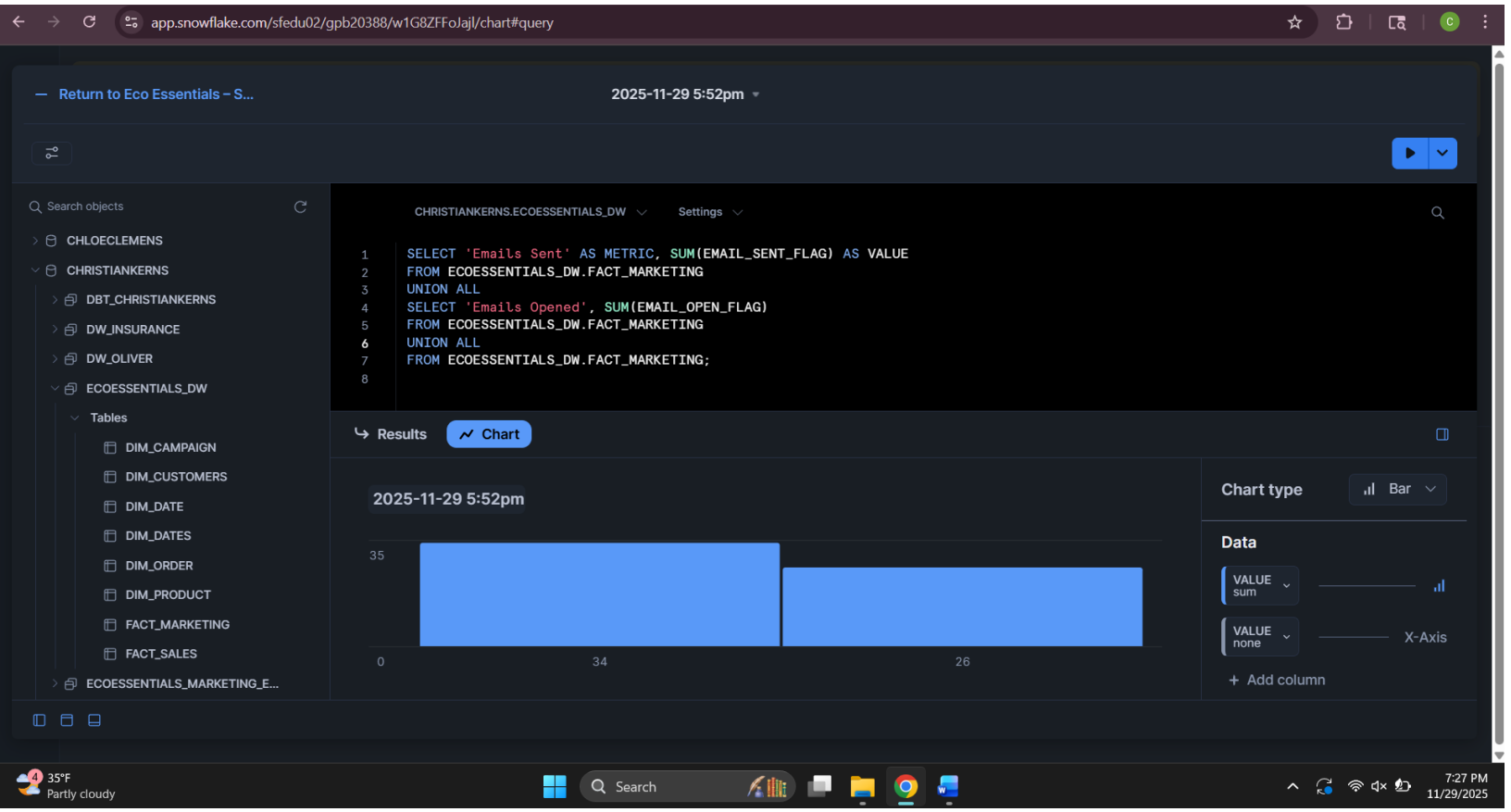
44°F Light rain Search 2:53 PM 11/16/2025

Completed project

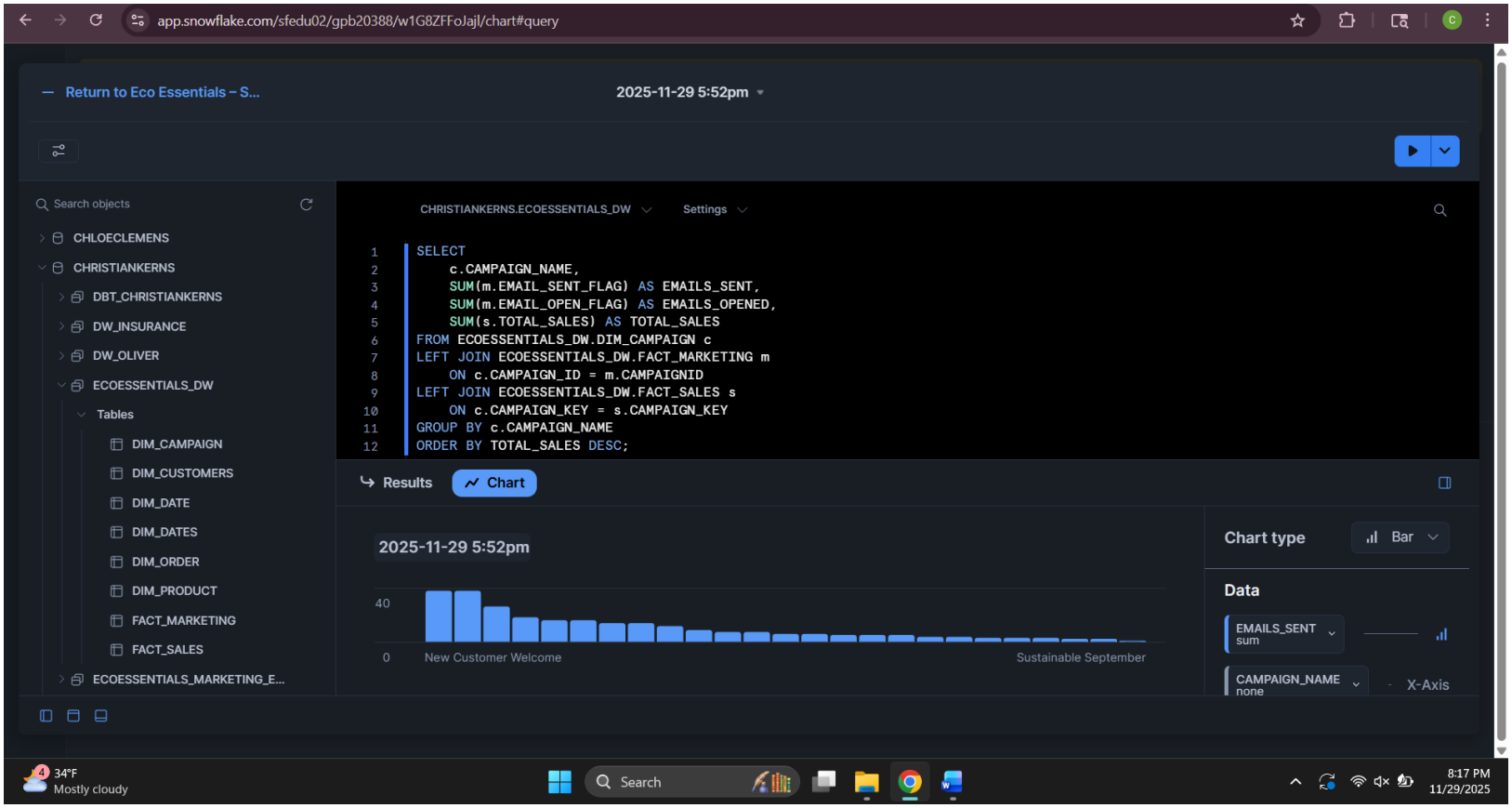
Sales Trend



Email Funnel



Campaign Performance



Explanation

Here we can see how Eco Essentials is doing with both sales and marketing. The sales trend shows how sales move up and down over time, so it's easy to spot when things are busy or slow. The email funnel breaks down how many emails were sent and how many people actually opened them, which helps show whether the marketing efforts are getting any attention. The campaign performance chart compares each campaign's email activity with the sales connected to it, making it clear which campaigns are actually helping bring in money. Altogether, the dashboard gives a quick, easy-to-understand picture of what's working and where there might be room to improve